Théa Ragon

Chargée de recherche, **CNRS**

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INVERSE PROBLEMS - SEISMIC SOURCE AND RUPTURE - SEISMIC CYCLE - FAULT SYSTEMS - GEODESY

ACADEMIC APPOINTMENTS

Jan. 2024 - present — Chargée de recherche CNRS · **ISTerre, Grenoble, France**
Assistant Professor equivalent, research position awarded by the French government (one of 5 positions awarded nationally across the Earth Sciences in 2023).

Sept. 2019 - Sept. 2023 — Post-doctoral Research Associate · **California Institute of Technology, Pasadena, U.S.A.**
2019-2021: Division of Geological and Planetary Sciences Chair's Prize Postdoctoral Fellowship.
Main collaborators: Mark Simons, Zachary Ross

March 2016 - Jul 2019 — Research assistant · **Géoazur, Université Côte d'Azur, France**
Uncertainties in earthquake source estimate, stemming from fault geometry, crustal structure, post-seismic deformation included in coseismic data. Advisors: A. Sladen and B. Delouis

2016 (2x6 months) — Junior Geologist · **IPGS- DYLBAS, Strasbourg and Total, Paris, France**
Rifting evolution of the Turkana Depression (Kenya) from seismic data. Topernawi Fm (Ekitale basin): sedimentary record of the earliest phase of rift initiation. Advisors: M. Schuster, A. Nutz, M. Allain, J.-L. Rubino.

EDUCATION

March 2016 - May 2019 — Ph.D. in Geophysics · **Géoazur, Université Côte d'Azur, France**

Sept. 2012 - 2015 — M.Sc in Geophysics & *Diplôme d'ingénieur* · **EOST, Université de Strasbourg, France**

Sept. 2010 - 2012 — CPGE · **Lycée Pierre de Fermat, Toulouse, France**

FELLOWSHIPS and AWARDS

2022 — AGU 2021 Editors' Citation for Excellence in Refereeing - JGR-Solid Earth

2019 - 2021 — Caltech GPS Chair's Prize Postdoctoral Fellowship.

2020 — Academic Excellence Fellowship and Thesis Award - Université Côte d'Azur

2016 - 2019 — French Ministery of Research and Higher Education Fellowship.

FUNDED OR SUCCESSFULL PROJECTS

2026-2028 — INSU Tellus, 25k€. PI T. Ragon. PICOR, Probabilistic Imaging of COntinental earthquake Ruptures: insights into fault zone properties.

2025-2028 — French Ministry PhD fellowship for Tom Hinton. main PI at 50%

2025 — CNES-APR GLiMe, 4.5k€. Co-PI

2024 — CNES-APR GLiMe, 9.5k€. Estimation du couplage intersismique et des interactions entre GLIssements lents et séismes dans la zone de subduction MExicaine en combinant données InSAR et GNSS et inférences stochastiques. Co-PI, PI Mathilde Radiguet.

2023, declined interview — CNES Postdoctoral Fellowship 2023. Discriminate distributed deformation from fault slips to unravel Mexican subduction behavior. PI T. Ragon.

2023, on reserve list — *MSCA* (Marie Skłodowska-Curie Actions) Postdoctoral Fellowship 2022. 3DiFFUSE, PI T. Ragon, with M. Radiguet (Isterre, Grenoble) and Y. van Dinther (Utrecht University). '*Seal of Excellence*': > 90% score.

TEACHING EXPERIENCE

2021. Université Côte d'Azur · total 10 hours, 20 students. LateX for PhD students & postdocs.

2016-2018. Université Côte d'Azur · total 142 hours, between 3 and ∼50 undergraduate students. Programming, Mathematics, Plate tectonics, Physics of the Earth, Field trips (cartography, seismic and electric acquisition).

ADVISING AND MENTORING

CSI - PhD advising committee — R. Tournier (2024-2027) - Hominin evolution in the east African rift.

PHD STUDENTS

Tom Hinton, **2025** - *2028* — *Probabilistic imaging of continental earthquake ruptures : insights into fault zone properties*. Advised at 50% with J. Hollingsworth and M.-L. Doan.

Leonid Pereiaslov, **2022** - *2027* — *Source inverse problems*. Caltech, co-advised at 20% with M. Simons.

ENGINEERS and JUNIOR SCIENTISTS

Islam Touzout, **2025-26** — 6 months. *Probabilistic estimates of plate interface coupling and uncertainties in the Mexican subduction zone from InSAR and GNSS*. Advised at 33%, in collaboration with M. Radiguet and E. Pathier.

MSC STUDENTS

Islam Touzout, **2025** — 6 months. *Probabilistic estimates of plate interface coupling and uncertainties in the Mexican subduction zone from InSAR and GNSS*. Advised at 50%, in collaboration with M. Radiguet and E. Pathier. 1 publication will result from this MSc project.

Emile Denise, **2024** — *Fonctions de Green empiriques : caractérisation des incertitudes et influence dans l'analyse de la source*. Advised at 90%, in collaboration with M. Causse. 1 publication in prep, but the student is facing personnal issues that will delay the submission.

Anne Maduel, **2024** — *GLiMe CNES project*. Co-advised at 15% with Mathilde Radiguet and Erwan Pathier.

Mathilde Banjan, **2017** — *Caractérisation du séisme d'Haida Gwaii (2012, Canada) avec des données tsunami et ionosphériques*, co-advised at 20% with L. Rolland and A. Sladen.

Training on supervision

2025, 16 hours — Optimisez votre communication et vos interactions doctorant – encadrant pendant la thèse, UGA

2025, 2 days — Formation encadrement doctoral, UGA

LEADERSHIP EXPERIENCE AND SCIENTIFIC COMMUNITY

Handling editor and **co-Chair of the Standards and Copy Editing Team** for **Seismica**, a community-led initiative for a Diamond Open Access journal for seismology and earthquake science.

Reviewer for scientific journals: Nature, Nature Communications, Science Advances, Earth Science Reviews, *Pure and Applied Geophysics* (PAAG), *Earth and Planetary Science Letters* (EPSL), Scientific Reports, *Bulletin of the Seismological Society of America* (BSSA), *Seismological Research Letters*, *Geophysical Journal International* (GJI), *Geophysical Research Letters* (GRL), *Journal of Geophysical Research* (JGR) Solid Earth, Tectonics, Geological Magazine, Computational Geosciences, Remote Sensing, *Comptes Rendus Geoscience*.

Reviewer for funding agencies: National Science Foundation (NSF, USA, one / year since 2022), National Fund for Scientific and Technological Development (FONDECYT, Chile, 2020 and 2022).

Organization of scientific meetings and seminars

2026 — Member of the Organizing Committee, 2nd School on Subduction zones, Cargèse, France.

Oct 2024 — Member of the Scientific Committee, 4th School on Earthquakes, Cargèse, France.

2024 - — In charge of the IRN Andes-Frensz webinar series, with V. Durand and N. Cerpa.

2024 - — In charge of seminars for the Cycle team, ISTerre.

2016 - 2019 — In charge of seminars for the Earthquake team, Géoazur.

International conference chair and convener

Dec. 2023, 2022, 2021, 2020, 2019 — Primary convener, or co-convener of seismology/source sessions at AGU Fall meeting.

Representative in several education and administrative boards

2025 - 2030 — Representative for college B researchers in the ISTerre laboratory committee.

2018 - 2019 — Student representative in the administrative board of Observatoire Côte d'Azur.

2017 - 2019 — Student representative in the Géoazur laboratory committee.

2013 - 2016 — Student representative in the board of Géophyse, EOST former students' association.

2012 - 2014 — Student representative in the EOST board of education.

2012 - 2013 — Secretary of the EOST (Strasbourg) student association.

Code development

I am involved in the development of several projects, including:

DeepGEM — PI T. Ragon, ISTerre. Bayesian inversion of source time functions and empirical Green's functions.

AlTar — PI M. Simons, Caltech. Bayesian inversion of slip.

CSI — PI R. Jolivet, ENS Paris. Finite-fault slip inversion

Pylith — v3 in particular, CIG, finite-element code for simulations of crustal deformation

PUBLICATIONS, SEMINARS, KEYNOTES & PRESENTATIONS

The full list is available **here**.